Solution

See Fig. 9.32.

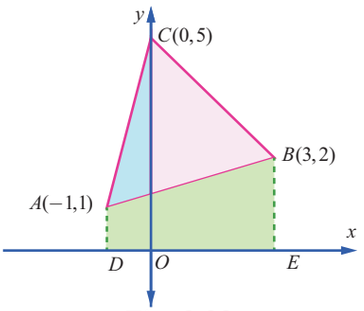


Fig. 9.32

Equation of AB is $\frac{y-1}{2-1}=\frac{x+1}{3+1}$ or $y=\frac{1}{4}(x+5)$

Equation of BC is $\frac{y-5}{2-5}=\frac{x-0}{3-0}$ or $y=-x+5$

Equation of AC is $\frac{y-1}{5-1}=\frac{x+1}{0+1}$ or $y=4x+5$

$\therefore$ Area of ΔABC = Area $DACO$+ Area of $OCBE$ − Area of $DABE$

$$= \int_{-1}^{0}(4x+5)\,dx+\int_{0}^{3}(-x+5)\,dx-\frac{1}{4}\int_{-1}^{3}(x+5)\,dx$$

$$= \left[\frac{4x^2}{2}+5x\right]_{-1}^{0}+\left[-\frac{x^2}{2}+5x\right]_{0}^{3}-\frac{1}{4}\left[\frac{x^2}{2}+5x\right]_{-1}^{3}$$

$$= 0-(+2-5)+\left(-\frac{9}{2}+15\right)-0-\frac{1}{4}\left[\frac{9}{2}+15\right]+\frac{1}{4}\left[\frac{1}{2}-5\right]=\frac{15}{2}$$

Example 9.61

Using integration, find the area of the region which is bounded by x-axis, the tangent and normal to the circle $x^2+y^2=4$ drawn at $\left(1,\sqrt{3}\right)$.

Solution

We recall that the equation of the tangent to the circle $x^2+y^2=a^2$ at (x_1,y_1) is $xx_1+yy_1=a^2$. So, the equation of the tangent to the circle $x^2+y^2=4$ at $\left(1,\sqrt{3}\right)$ is $x+y\sqrt{3}=4$; that is, $y=-\frac{1}{\sqrt{3}}(x-4)$. The tangent meets the x-axis at the point (4,0).

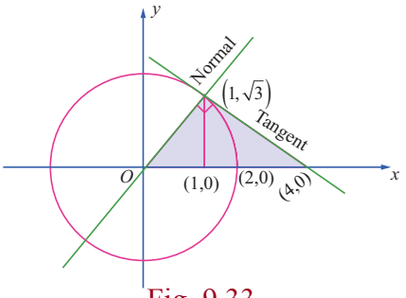


Fig. 9.33

The slope of the tangent is $-\frac{1}{\sqrt{3}}$. So the slope of the normal is $\sqrt{3}$ and hence equation of the normal is $y-\sqrt{3}=\sqrt{3}(x-1)$; that is $y=\sqrt{3}x$ and it passes through the origin. The area to be found is shaded in the adjoining figure. It can be found by two methods.

Method 1

Viewing in the postive direction of y-axis, the required area is the area of the region bounded by x-axis, $y=\sqrt{3}x$ and $x+y\sqrt{3}=4$. So it can be obtained by applying the formula $\int_a^b y\,dx$. For this, we have to split the region into sub-regions, one sub-region bounded by x-axis, the normal $y=\sqrt{3}x$ and the line $x=1$; the other sub-region bounded by x-axis, the tangent $x+y\sqrt{3}=4$ and the line $x=1$ axis.

$$\therefore \text{ Area required } = \int_0^1 y\,dx+\int_1^4 y\,dx=\int_0^1 \sqrt{3}x\,dx+\int_1^4\left[-\frac{1}{\sqrt{3}}(x-4)\right]dx$$

$$= \left[\sqrt{3}\frac{x^2}{2}\right]_0^1+\left[-\frac{1}{\sqrt{3}}\left(\frac{x^2}{2}-4x\right)\right]_1^4=\frac{\sqrt{3}}{2}+\frac{8}{\sqrt{3}}-\frac{7}{2\sqrt{3}}=2\sqrt{3}\,.$$

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